

Checklist B: Communication Entropy Tally Sheet

1. Purpose and Scope

This instrument measures Communication Entropy (CE), defined as the ratio of discrete communication artefacts to successful technical actions executed during a defined response phase. CE operationalises the phenomenon of coordination overhead, distinguishing productive communication that supports action from excessive or unstructured communication that replaces it.

A high CE ratio indicates that teams are negotiating each response step in real time rather than executing pre-authorised procedures. This pattern is diagnostic of absent decision triggers, unclear role boundaries, and insufficient IRP operationalisation rather than a reflection of individual communication ability.

This sheet is completed by the designated evaluator during each defined response phase. It informs the following IRRS sub-metrics: Communication logging, Decision flow, and Role clarity.

2. Pre-Simulation Setup

Complete the following fields before the simulation begins.

Organisation / Team	
Simulation Date	
Scenario Class	<i>Ransomware / Insider Threat / Cloud Misconfiguration / Credential Leakage (circle one)</i>
SRI Level	<i>1 / 2 / 3 / 4 / 5 (circle one)</i>
Evaluator Name	

3. Counting Rules

Accuracy in message counting is essential for a reliable CE ratio. Apply the following rules consistently across all phases.

Count these (M total) Chat messages sent between team members Verbal exchanges and spoken directives Questions posed to other team members Requests for confirmation or approval Formal escalation communications	Do not count these Facilitator inject announcements Evaluator prompts or observer questions Post-exercise debrief discussion Technical system alerts and automated notifications Communications outside the active response phase
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4. Communication Pattern Classification

Before recording phase data, familiarise yourself with the three communication patterns below. Record the dominant pattern observed during each phase in the tally sheet.

Pattern Label	Typical CE Range	Description
Instruction-driven	Under 10:1	Clear role structure and pre-authorised decision triggers. Communication is concise and action-oriented. Roles are well understood and team executes without requiring consensus.
Consensus seeking	10:1 to 20:1	Team discusses options before acting. Decision authority is present but not consistently exercised. Response is slower than optimal but retains a degree of structure.
Approval seeking	Over 20:1	Team negotiates each step in real time, seeking permission rather than executing predefined protocols. Indicates absence of pre-authorised decision triggers and significant role clarity deficit.

5. Communication Entropy Tally Sheet

Record one row per active response phase. M total is the count of discrete messages or utterances. A valid is the count of successful technical response actions executed during the phase. CE Ratio is M total divided by A valid. Communication Pattern is selected from the classification table above.

Phase	M total (messages)	A valid (actions)	CE Ratio (M/A)	Communication Pattern	Notes
Triage					
Containment					
Eradication					
Recovery					
CE Range across all phases:			to	Overall Pattern Observed:	

6. Scenario Threshold Guidance

The following reference bands indicate when a calculated CE ratio should be considered acceptable, elevated, or critical within a given scenario class. Higher entropy thresholds are applied to lower SRI scenarios because investigative and governance-oriented incidents naturally involve greater deliberation before action.

Important: A CE ratio exceeding 20:1 in a ransomware scenario (SRI 5) is a strong indicator that teams are negotiating containment steps rather than executing pre-authorised protocols and should be classified as critical. In an insider threat scenario (SRI 4), a moderately higher ratio may reflect appropriate caution given the governance sensitivity of the incident, and the threshold for critical classification is therefore set higher. Always interpret CE ratios relative to the scenario class and SRI level recorded in Section 2.

Scenario Class	SRI	Acceptable	Elevated	Critical
Ransomware Propagation	SRI 5	Under 10:1	10:1 to 20:1	Over 20:1
Insider Data Exfiltration	SRI 4	Under 15:1	15:1 to 25:1	Over 25:1

Cloud Misconfiguration	SRI 3	Under 20:1	20:1 to 30:1	Over 30:1
Credential Leakage	SRI 3	Under 20:1	20:1 to 30:1	Over 30:1

7. Worked Example (Insider Threat, SRI 4)

The following completed example is drawn from the theoretical worked example presented in the IRRS paper.

Phase	M total	A valid	CE Ratio	Pattern	Notes
Triage	46	2	23.0	Consensus seeking	Team debated scope of the incident before any containment action was issued
Containment	78	3	26.0	Approval seeking	Multiple requests for managerial sign-off before each technical action; no pre-authorised decision triggers in place
CE Range: 23.0 to 26.0 Classification: ELEVATED to CRITICAL (SRI 4)			<i>Dominant pattern: teams negotiated each step in real time rather than executing pre-authorised protocols</i>		

The CE range of 23.0 to 26.0 across both phases falls within the Elevated to Critical band for an SRI 4 scenario. The dominant pattern of approval seeking across both phases indicates an absence of pre-authorised decision triggers rather than a lack of technical knowledge. This directly explains the moderate scores in the Communication logging (si = 3, wi = 4) and Decision flow (si = 3, wi = 5) sub-metrics in the corresponding IRRS calculation.

8. Evaluator Guidance Notes

- Define phase boundaries clearly before the simulation begins. Triage typically ends when the incident is confirmed and containment begins. Agree these boundaries with the simulation facilitator in advance.
- Where two phases overlap or transition rapidly, record the boundary timestamp and split counts proportionally based on the facilitator inject log.
- If a team takes no valid actions during a phase, record A valid as zero and note the absence of action. Do not calculate a CE ratio for that phase as division by zero is undefined; instead, record the phase as producing no measurable throughput.
- Record the dominant communication pattern per phase rather than attempting to categorise individual messages. The pattern reflects the overall coordination dynamic of the team during that phase.
- This instrument is designed for system-level diagnostic analysis only and must not be used to evaluate individual communication ability or interpersonal behaviour.

This instrument is part of the IRRS Human-Centric Telemetry suite. It should be used alongside Checklist A (Decision Latency Capture Sheet) and Checklist C (Authority Drift Tracker) to provide a complete diagnostic picture of human-system interaction during the simulation.